Git & GitHub Commands — Common Situations

Practical command sheet for Windows CMD / Git Bash. Use the section that matches your situation.

1. First Time: Push an Existing Local Project to a NEW GitHub Repository

Situation

```
You already have a project folder on your PC and created an empty repository on GitHub.
```

Commands

```
cd "C:\path\to\your\project"
git init
git add .
git commit -m "Initial commit"
git branch -M main
git remote add origin https://github.com/USERNAME/REPOSITORY.git
git push -u origin main
```

Example for your current project

```
cd "C:\Users\adity\OneDrive\Desktop\project Config service folder"
git init
git add .
git commit -m "Initial commit"
git branch -M main
git remote add origin https://github.com/AdityaZ11002/project-Config-service-folder.git
git push -u origin main
```

2. GitHub Repository Already Has Files / You Get a Non-Fast-Forward Error

Situation

```
The GitHub repository already contains a README, .gitignore, or another commit.
```

Commands

```
git pull origin main --rebase
git push -u origin main
```

If Git reports unrelated histories

```
git pull origin main --allow-unrelated-histories
git push -u origin main
```

3. Normal Update: You Changed Existing Project Files

Situation

```
Your project is already connected to GitHub and you want to upload your latest changes.
```

Commands

```
git status
git add .
git commit -m "Updated project"
git push
```

4. Check What Changed Before Committing

Commands

```
git status
```

```
git diff
```

Useful

```
git status shows modified/untracked files. git diff shows the actual line changes.
```

5. Add Only Specific Files

Commands

```
git add filename.java
git add src/main/java/com/example/App.java
git commit -m "Updated specific files"
git push
```

6. Undo git add Before Commit

Situation

```
You ran git add but have not committed yet.
```

Commands

```
git restore --staged .
git status
```

For one file

```
git restore --staged filename.java
```

7. Change the Last Commit Message

Commands

```
git commit --amend -m "Correct commit message
```

Warning

```
Use carefully if the old commit has already been pushed.
```

8. Check Remote GitHub URL

Commands

```
git remote -v
```

Change Remote URL

```
git remote set-url origin https://github.com/USERNAME/REPOSITORY.git
```

9. Remote 'origin' Already Exists

Situation

```
You run git remote add origin ... and Git says origin already exists.
```

Commands

```
git remote -v
git remote set-url origin https://github.com/USERNAME/REPOSITORY.git
git push -u origin main
```

10. Rename Branch to main

Commands

```
git branch -M main
git push -u origin main
```

11. Create a New Branch

Commands

```
git switch -c feature-login
git push -u origin feature-login
```

Go back to main

```
git switch main
```

12. See All Branches

Commands

```
git branch
git branch -a
```

13. Merge a Branch into main

Commands

```
git switch main
git pull origin main
git merge feature-login
git push origin main
```

14. Delete a Local Branch

Commands

```
git branch -d feature-login
```

Force delete

```
git branch -D feature-login
```

15. Delete a Remote Branch

Commands

```
git push origin --delete feature-login
```

16. Clone an Existing GitHub Repository

Situation

```
The project already exists on GitHub and you want a copy on your PC.
```

Commands

```
cd "C:\Users\adity\OneDrive\Desktop"
git clone https://github.com/USERNAME/REPOSITORY.git
cd REPOSITORY
```

17. Pull Latest Changes from GitHub

Commands

```
git pull origin main
```

18. Fetch Without Changing Your Working Files

Commands

```
git fetch origin
git status
git log --oneline --all
```

19. View Commit History

Commands

```
git log --oneline
git log --oneline --all --graph
```

20. Undo the Last Local Commit (Keep Changes)

Situation

```
You committed locally but want to edit the files and commit again.
```

Commands

```
git reset --soft HEAD~1
```

21. Undo the Last Local Commit and Discard Changes

Situation

```
Only use this when you are sure you do not need those changes.
```

Commands

```
git reset --hard HEAD~1
```

Warning

```
This can permanently discard uncommitted work.
```

22. Restore a Deleted/Changed File

Commands

```
git restore filename.java
```

Restore all unstaged changes

```
git restore .
```

23. Create a .gitignore for Java / Maven / Spring Boot

Situation

```
You do not want IDE files, Maven target files, logs, or secrets in GitHub.
```

Example .gitignore

```
.idea/
*.iml
.classpath
.project
.settings/
target/
*.log
.env
application-local.properties
```

Then

```
git add .
git commit -m "Add gitignore"
git push
```

24. Check Git Installation / User Configuration

Commands

```
git --version
git config --global user.name
git config --global user.email
```

Set identity

```
git config --global user.name "Your Name"
git config --global user.email "your-email@example.com"
```

25. Authentication / Login Problem

Situation

```
GitHub may reject password-based HTTPS authentication. Use GitHub authentication, Git Credential Manager,
```

Useful check

```
git remote -v
git config --global credential.helper
```

26. Git Push Says 'Everything Up-to-Date'

Commands

```
git status
git log --oneline -3
git remote -v
```

Meaning

```
Usually there are no new committed changes to send, or you are pushing a different branch than expected.
```

27. Accidentally Added a Secret / Password

Immediate action

```
Do not simply rely on deleting the file in a later commit. Revoke/rotate the exposed secret first, then re
```

Prevention

```
echo application-local.properties >> .gitignore
echo .env >> .gitignore
```

28. Quick Daily Workflow

Commands

```
git status
git pull
git add .
git commit -m "Describe your changes"
git push
```

Quick Decision Guide

If you want to…	Use
Push a local folder to a new empty GitHub repo	git init → git add . → git commit → git remote add → git push
Upload normal new changes	git add . → git commit → git push
Get latest GitHub changes	git pull origin main
Create a feature branch	git switch -c feature-name
Merge feature into main	git switch main → git merge feature-name → git push
See what changed	git status / git diff
Fix 'origin already exists'	git remote set-url origin ...
Check GitHub connection	git remote -v

Important: Replace USERNAME and REPOSITORY with your own GitHub values. Before using destructive commands such as reset --hard, make sure you do not need the changes.